**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

DAIRY FACILITY INFORMATION

**A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY:** Vitorino Dairy (Caetano Dairy Facility)

Physical address of dairy:

| 9436 Griffith AVE | Delhi | Merced | 95315 |
|---|---|---|---|
| Number and Street | City | County | Zip Code |

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1962

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X046-X120-X001-XXXX

**B. OPERATORS**

Caetano, Paul

Operator name: Caetano, Paul

Telephone no.: (209) 632-4162 (Landline) (Cellular)

| 9436 Griffith AVE | Delhi | CA | 95315 |
|---|---|---|---|
| Mailing Address Number and Street | City | State | Zip Code |

Vitorino, John

Operator name: Vitorino, John

Telephone no.: (209) 535-6259 (Landline) (Cellular)

| 3918 S Kilroy RD | Turlock | CA | 95380 |
|---|---|---|---|
| Mailing Address Number and Street | City | State | Zip Code |

**This operator is responsible for paying permit fees.**

**C. OWNERS**

Caetano, Paul

Legal owner name: Caetano, Paul

Telephone no.: (209) 632-4162 (Landline) (Cellular)

| 9436 Griffith AVE | Delhi | CA | 95315 |
|---|---|---|---|
| Mailing Address Number and Street | City | State | Zip Code |

**Annual Report - General Order No. R5-2007-0035**
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AVAILABLE NUTRIENTS

## A. HERD INFORMATION

| | Milk Cows | Dry Cows | Bred Heifers (15-24 mo.) | Heifers (7-14 mo. to breeding) | Calves (4-6 mo.) | Calves (0-3 mo.) |
|---|---|---|---|---|---|---|
| Number open confinement | 0 | 0 | 0 | 0 | 0 | 0 |
| Number under roof | 635 | 50 | 0 | 0 | 205 | 0 |
| Maximum number | 635 | 50 | 0 | 0 | 205 | 0 |
| Average number | 635 | 50 | 0 | 0 | 205 | 0 |
| Avg live weight (lbs) | 1,500 | 1,300 | 0 | 0 | | |

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 73 *pounds per cow per day*

## B. MANURE GENERATED

Total manure excreted by the herd: 17,924.89 *tons per reporting period*

Total nitrogen from manure: 235,490.01 *lbs per reporting period*

After ammonia losses (30% loss applied): 164,843.01 *lbs per reporting period*

Total phosphorus from manure: 38,536.68 *lbs per reporting period*

Total potassium from manure: 122,859.94 *lbs per reporting period*

Total salt from manure: 310,487.25 *lbs per reporting period*

## C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 16,758,000 *gallons*

Total nitrogen generated: 48,982.90 *lbs*

Total phosphorus generated: 9,729.70 *lbs*

Total potassium generated: 71,403.00 *lbs*

Total salt generated: 446,552.55 *lbs*

| | | |
|---|---|---|
| | 24,962,782 | *gallons applied* |
| + | 0 | *gallons exported* |
| - | 8,204,782 | *gallons imported* |
| = | 16,758,000 | *gallons generated* |

## D. FRESH WATER SOURCES

| Source Description | Type |
|---|---|
| Domestic Well - 1-B 9436 Griffith | Ground water |
| Domestic Well - 2-H 9436 Griffith | Ground water |
| Domestic Well - 4-17573 Schendel | Ground water |
| Irrigation Water - Canal | Surface water |

## E. SUBSURFACE (TILE) DRAINAGE SOURCES

*No subsurface (tile) drainage sources entered.*

## F. NUTRIENT IMPORTS

| Date | Material type / Description | Quantity | Reporting basis | Moist. (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/28/2023 | Corral solids<br>*SMI - WJF* | 650.00 *ton* | Dry-weight | 33.1 | 12,000.00 | 3,800.00 | 13,700.00 | | 76.20 |
| 10/16/2023 | Corral solids<br>*SMI - WJF* | 800.00 *ton* | Dry-weight | 18.1 | 4,100.00 | 2,800.00 | 14,700.00 | | 88.10 |

| Date | Material type / Description | Quantity | N (mg/L) | P (mg/L) | K (mg/L) | Salt (mg/L) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|
| 03/24/2023 | Process wastewater<br>*LWI - WJF* | 2,320,101.00 *gal* | 314.00 | 0.00 | 961.00 | | 4,120 |
| 05/19/2023 | Process wastewater<br>*LWI - WJF* | 1,089,230.00 *gal* | 627.00 | 0.00 | 996.00 | | 6,340 |
| 10/22/2023 | Process wastewater<br>*LWI - WJF* | 4,795,451.00 *gal* | 123.00 | 0.00 | 362.00 | | 3,480 |

*No commercial or other nutrient imports entered.*

| Material type | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 15,809.04 | 6,973.98 | 31,177.77 | 1,817,173.80 |
| Process wastewater | 16,700.84 | 0.00 | 42,145.95 | 276,659.31 |
| Total imports for all materials | 32,509.88 | 6,973.98 | 73,323.72 | 2,093,833.11 |

## G. NUTRIENT EXPORTS

*No solid nutrient exports entered.*

*No liquid nutrient exports entered.*

APPLICATION AREA

## A. LIST OF LAND APPLICATION AREAS

| Field name | Controlled acres | Cropable acres | Total harvests | Type of waste applied | Parcel number |
|---|---|---|---|---|---|
| Field 1 | 28 | 28 | 0 | none | X046-X110-X035-XXXX |
| Field 2 | 10 | 10 | 2 | process wastewater | X046-X120-X001-XXXX |
| Field 3 | 20 | 20 | 2 | process wastewater | X046-X120-X002-XXXX |
| Field 4 | 4 | 4 | 0 | none | X046-X120-X001-XXXX |
| Field 5 | 17 | 17 | 2 | process wastewater | X046-X120-X002-XXXX |
| Field 6 | 11 | 11 | 2 | process wastewater | X045-X150-X050-XXXX |
| Field 7 | 30 | 30 | 2 | process wastewater | X045-X150-X050-XXXX |
| HR-East | 41 | 41 | 0 | none | X045-X240-X036-XXXX |
| HR-North | 82 | 82 | 0 | none | X045-X240-X036-XXXX |
| HR-South | 82 | 82 | 2 | both | X045-X251-X002-XXXX |
| Totals for areas that were used for application | 170 | 170 | 12 | | |
| Totals for areas that were not used for application | 155 | 155 | 0 | | |
| Land application area totals | 325 | 325 | 12 | | |

## B. CROPS AND HARVESTS

Field 2

Field name: Field 2

12/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 12/08/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/15/2023 | 110.50 *ton* | Dry-weight | | 60.7 | 20,300.00 | 4,100.00 | 34,000.00 | | 17.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 11.05 | 176.31 | 35.61 | 295.30 | 1,545.98 |

## Field 2

### 05/30/2023: Corn, silage

Crop: Corn, silage  Acres planted: 10  Plant date: 05/30/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/14/2023 | 260.20 *ton* | Dry-weight | | 58.8 | 12,400.00 | 2,500.00 | 8,300.00 | | 5.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.02 | 265.86 | 53.60 | 177.96 | 1,157.79 |

## Field 3

Field name: Field 3

### 12/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 20  Plant date: 12/08/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/15/2023 | 204.50 *ton* | Dry-weight | | 60.7 | 20,300.00 | 4,100.00 | 34,000.00 | | 17.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 10.23 | 163.15 | 32.95 | 273.25 | 1,430.56 |

### 05/15/2023: Corn, silage

Crop: Corn, silage  Acres planted: 20  Plant date: 05/15/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/14/2023 | 532.60 *ton* | Dry-weight | | 58.8 | 12,400.00 | 2,500.00 | 8,300.00 | | 5.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.63 | 272.09 | 54.86 | 182.13 | 1,184.93 |

## Field 5

Field name: Field 5

### 12/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 17  Plant date: 12/08/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/15/2023 | 192.60 *ton* | Dry-weight | | 66.5 | 19,700.00 | 4,200.00 | 34,200.00 | | 12.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 11.33 | 149.54 | 31.88 | 259.60 | 964.02 |

### 05/15/2023: Corn, silage

Crop: Corn, silage  Acres planted: 17  Plant date: 05/15/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/14/2023 | 448.90 *ton* | Dry-weight | | 57.8 | 11,900.00 | 2,300.00 | 7,700.00 | | 5.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.41 | 265.21 | 51.26 | 171.61 | 1,114.33 |

## Field 6

Field name: Field 6

### 12/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 11  Plant date: 12/08/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/15/2023 | 120.40 *ton* | Dry-weight | | 64.0 | 20,200.00 | 4,300.00 | 34,200.00 | | 16.20 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 10.95 | 159.19 | 33.89 | 269.52 | 1,276.68 |

## Field 6

### 05/15/2023: Corn, silage

Crop: Corn, silage  Acres planted: 11  Plant date: 05/15/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/14/2023 | 290.60 *ton* | Dry-weight | | 58.5 | 11,800.00 | 2,300.00 | 7,600.00 | | 4.90 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.42 | 258.74 | 50.43 | 166.65 | 1,074.43 |

## Field 7

Field name: Field 7

### 12/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 30  Plant date: 12/08/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/15/2023 | 331.50 *ton* | Dry-weight | | 65.6 | 20,000.00 | 4,300.00 | 34,900.00 | | 13.30 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 11.05 | 152.05 | 32.69 | 265.32 | 1,011.12 |

### 05/15/2023: Corn, silage

Crop: Corn, silage  Acres planted: 30  Plant date: 05/15/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/14/2023 | 790.20 *ton* | Dry-weight | | 58.4 | 12,100.00 | 2,400.00 | 7,700.00 | | 5.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.34 | 265.17 | 52.60 | 168.74 | 1,095.74 |

## HR-South

Field name: HR-South

### 12/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 82  Plant date: 12/08/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/15/2023 | 968.60 *ton* | Dry-weight | | 65.8 | 20,700.00 | 4,400.00 | 35,600.00 | | 12.90 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 11.81 | 167.25 | 35.55 | 287.63 | 1,042.26 |

### 05/15/2023: Corn, silage

Crop: Corn, silage  Acres planted: 82  Plant date: 05/15/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/14/2023 | 2,055.60 *ton* | Dry-weight | | 58.5 | 11,700.00 | 2,200.00 | 7,200.00 | | 4.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 25.07 | 243.44 | 45.77 | 149.81 | 977.91 |

NUTRIENT BUDGET

## A. LAND APPLICATIONS

### Field 2 - 12/08/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough — Plant date: 12/08/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/07/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 163.15 | 23.70 | 205.33 | 995.89 | 351,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 37.09 | 1,481,700.00 *gal* |
| Application event totals | | 163.15 | 23.70 | 205.33 | 1,032.99 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 74.39 | 19.40 | 119.21 | 911.47 | 333,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.25 | 1,448,025.00 *gal* |
| Application event totals | | 74.39 | 19.40 | 119.21 | 947.73 | |

### Field 2 - 05/30/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage — Plant date: 05/30/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 42.15 | 1,683,750.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 42.15 | |

## Field 2 - 05/30/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.47 | 1,616,400.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 40.47 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 134.18 | 18.57 | 182.78 | 840.65 | 351,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.00 | 1,717,425.00 *gal* |
| Application event totals | | 134.18 | 18.57 | 182.78 | 883.65 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 137.62 | 19.05 | 187.46 | 862.21 | 360,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 41.31 | 1,650,075.00 *gal* |
| Application event totals | | 137.62 | 19.05 | 187.46 | 903.52 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 82.43 | 21.49 | 132.10 | 1,010.01 | 369,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.00 | 1,717,425.00 *gal* |
| Application event totals | | 82.43 | 21.49 | 132.10 | 1,053.01 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 41.31 | 1,650,075.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 41.31 | |

## Field 2 - 05/30/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.00 | 1,717,425.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 43.00 | |

## Field 3 - 12/08/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 12/08/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/07/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 73.39 | 19.13 | 117.60 | 899.16 | 657,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.36 | 3,064,425.00 *gal* |
| Application event totals | | 73.39 | 19.13 | 117.60 | 937.52 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 140.14 | 20.35 | 176.37 | 855.45 | 603,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.67 | 2,929,725.00 *gal* |
| Application event totals | | 140.14 | 20.35 | 176.37 | 892.12 | |

## Field 3 - 05/15/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/15/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

## Field 3 - 05/15/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.36 | 3,064,425.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 38.36 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.62 | 3,165,450.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.62 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.67 | 2,929,725.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 36.67 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 91.48 | 23.85 | 146.60 | 1,120.87 | 819,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.36 | 3,064,425.00 *gal* |
| Application event totals | | 91.48 | 23.85 | 146.60 | 1,159.23 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 87.46 | 22.80 | 140.16 | 1,071.60 | 783,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 37.52 | 2,997,075.00 *gal* |
| Application event totals | | 87.46 | 22.80 | 140.16 | 1,109.11 | |

## Field 3 - 05/15/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 177.79 | 25.82 | 223.76 | 1,085.27 | 765,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.67 | 2,929,725.00 *gal* |
| Application event totals | | 177.79 | 25.82 | 223.76 | 1,121.94 | |

| | | | | |
|---|---|---|---|---|
| 08/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.36 | 3,064,425.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 38.36 | |

## Field 5 - 12/08/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough Plant date: 12/08/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/07/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 94.61 | 24.67 | 151.62 | 1,159.27 | 720,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.68 | 2,626,650.00 *gal* |
| Application event totals | | 94.61 | 24.67 | 151.62 | 1,197.95 | |

| | | | | |
|---|---|---|---|---|
| 04/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 93.43 | 24.36 | 149.73 | 1,144.78 | 711,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.19 | 2,592,975.00 *gal* |
| Application event totals | | 93.43 | 24.36 | 149.73 | 1,182.96 | |

## Field 5 - 05/15/2023: Corn, silage

## Field 5 - 05/15/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 05/15/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.14 | 2,929,725.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 43.14 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.18 | 2,660,325.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.18 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 165.96 | 22.97 | 226.06 | 1,039.72 | 738,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.19 | 2,592,975.00 *gal* |
| Application event totals | | 165.96 | 22.97 | 226.06 | 1,077.90 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 91.07 | 23.74 | 145.94 | 1,115.80 | 693,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.18 | 2,660,325.00 *gal* |
| Application event totals | | 91.07 | 23.74 | 145.94 | 1,154.97 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 88.70 | 23.13 | 142.15 | 1,086.81 | 675,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.19 | 2,592,975.00 *gal* |
| Application event totals | | 88.70 | 23.13 | 142.15 | 1,125.00 | |

## Field 5 - 05/15/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.14 | 2,929,725.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 43.14 | |

| | | | | |
|---|---|---|---|---|
| 08/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.18 | 2,660,325.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.18 | |

## Field 6 - 12/08/2022: Oats, silage-soft dough

Field name: Field 6

Crop: Oats, silage-soft dough

Plant date: 12/08/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/06/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 140.71 | 20.44 | 177.09 | 858.93 | 333,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.85 | 1,751,100.00 *gal* |
| Application event totals | | 140.71 | 20.44 | 177.09 | 898.78 | |

| | | | | |
|---|---|---|---|---|
| 03/31/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 71.28 | 18.59 | 114.23 | 873.40 | 351,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.62 | 1,784,775.00 *gal* |
| Application event totals | | 71.28 | 18.59 | 114.23 | 914.02 | |

## Field 6 - 05/15/2023: Corn, silage

## Field 6 - 05/15/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 05/15/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.32 | 1,683,750.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 38.32 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.62 | 1,784,775.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 40.62 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 100.53 | 26.21 | 161.10 | 1,231.72 | 495,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 42.15 | 1,852,125.00 *gal* |
| Application event totals | | 100.53 | 26.21 | 161.10 | 1,273.87 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 96.87 | 25.26 | 155.24 | 1,186.93 | 477,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.09 | 1,717,425.00 *gal* |
| Application event totals | | 96.87 | 25.26 | 155.24 | 1,226.02 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 147.01 | 20.35 | 200.24 | 920.99 | 423,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.85 | 1,751,100.00 *gal* |
| Application event totals | | 147.01 | 20.35 | 200.24 | 960.85 | |

## Field 6 - 05/15/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.09 | 1,717,425.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.09 | |

| | | | | |
|---|---|---|---|---|
| 08/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 42.15 | 1,852,125.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 42.15 | |

## Field 7 - 12/08/2022: Oats, silage-soft dough

Field name: Field 7

Crop: Oats, silage-soft dough Plant date: 12/08/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/06/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 128.45 | 17.78 | 174.96 | 804.73 | 1,008,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 31.19 | 3,737,925.00 *gal* |
| Application event totals | | 128.45 | 17.78 | 174.96 | 835.92 | |

| | | | | |
|---|---|---|---|---|
| 03/31/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 75.73 | 19.75 | 121.36 | 927.90 | 1,017,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 32.32 | 3,872,625.00 *gal* |
| Application event totals | | 75.73 | 19.75 | 121.36 | 960.21 | |

## Field 7 - 05/15/2023: Corn, silage

## Field 7 - 05/15/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage

Plant date: 05/15/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 41.31 | 4,950,225.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 41.31 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 41.87 | 5,017,575.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 41.87 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 89.81 | 23.42 | 143.92 | 1,100.34 | 1,206,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.47 | 4,849,200.00 *gal* |
| Application event totals | | 89.81 | 23.42 | 143.92 | 1,140.80 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 161.71 | 22.38 | 220.27 | 1,013.09 | 1,269,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 41.31 | 4,950,225.00 *gal* |
| Application event totals | | 161.71 | 22.38 | 220.27 | 1,054.40 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 93.16 | 24.29 | 149.29 | 1,141.40 | 1,251,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 42.43 | 5,084,925.00 *gal* |
| Application event totals | | 93.16 | 24.29 | 149.29 | 1,183.83 | |

## Field 7 - 05/15/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 37.94 | 4,546,125.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 37.94 | |

| | | | | |
|---|---|---|---|---|
| 08/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.50 | 4,613,475.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 38.50 | |

## HR-South - 12/08/2022: Oats, silage-soft dough

Field name: HR-South

Crop: Oats, silage-soft dough

Plant date: 12/08/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/06/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.79 | 13,032,225.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.79 | |

| | | | | |
|---|---|---|---|---|
| 03/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - WJF | Process wastewater | 74.14 | 0.00 | 226.90 | 972.78 | 2,320,101.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.99 | 13,099,575.00 *gal* |
| Application event totals | | 74.14 | 0.00 | 226.90 | 1,012.78 | |

| | | | | |
|---|---|---|---|---|
| 10/16/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure - Import WJF | Corral solids | 127.27 | 40.30 | 145.30 | 8,081.85 | 650.00 *ton* |
| Application event totals | | 127.27 | 40.30 | 145.30 | 8,081.85 | |

## HR-South - 12/08/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - WJF | Process wastewater | 26.12 | 0.00 | 76.87 | 738.96 | 2,086,543.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.10 | 13,133,250.00 *gal* |
| Application event totals | | 26.12 | 0.00 | 76.87 | 779.05 | |

## HR-South - 05/15/2023: Corn, silage

Field name: HR-South

Crop: Corn, silage

Plant date: 05/15/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure - Import WJF | Corral solids | 65.52 | 44.75 | 234.91 | 14,078.81 | 800.00 *ton* |
| Solid Manure | Corral solids | 136.52 | 49.64 | 68.83 | 0.00 | 600.00 *ton* |
| Application event totals | | 202.04 | 94.39 | 303.74 | 14,078.81 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - WJF | Process wastewater | 69.50 | 0.00 | 110.41 | 702.78 | 1,089,230.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.58 | 12,964,875.00 *gal* |
| Application event totals | | 69.50 | 0.00 | 110.41 | 742.37 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.99 | 13,099,575.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.99 | |

HR-South - 05/15/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - WJF | Process wastewater | 18.30 | 0.00 | 53.87 | 517.82 | 1,462,130.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.20 | 13,166,925.00 *gal* |
| Application event totals | | 18.30 | 0.00 | 53.87 | 558.02 | |

| | | | | |
|---|---|---|---|---|
| 07/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.79 | 13,032,225.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.79 | |

| | | | | |
|---|---|---|---|---|
| 08/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.58 | 12,964,875.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.58 | |

| | | | | |
|---|---|---|---|---|
| 08/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.99 | 13,099,575.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.99 | |

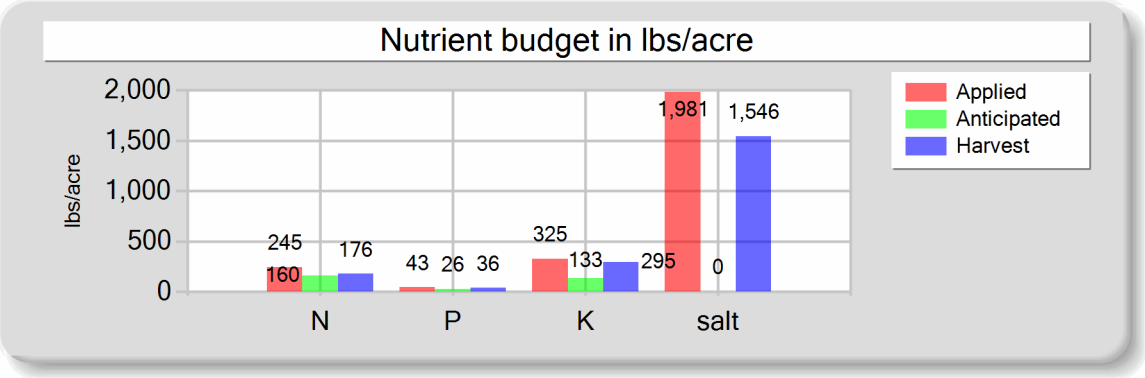
| | | | | |
|---|---|---|---|---|
| 08/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - WJF | Process wastewater | 15.61 | 0.00 | 45.93 | 441.55 | 1,246,778.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.10 | 13,133,250.00 *gal* |
| Application event totals | | 15.61 | 0.00 | 45.93 | 481.65 | |

## B. NUTRIENT BUDGET

Field 2 - 12/08/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 12/08/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 237.54 | 43.09 | 324.54 | 1,907.37 |
| Fresh water | 0.00 | 0.00 | 0.00 | 73.35 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 244.54 | 43.09 | 324.54 | 1,980.71 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 176.31 | 35.61 | 295.30 | 1,545.98 |
| Nutrient balance | 68.23 | 7.48 | 29.24 | 434.73 |
| Applied to removed ratio | 1.39 | 1.21 | 1.10 | 1.28 |

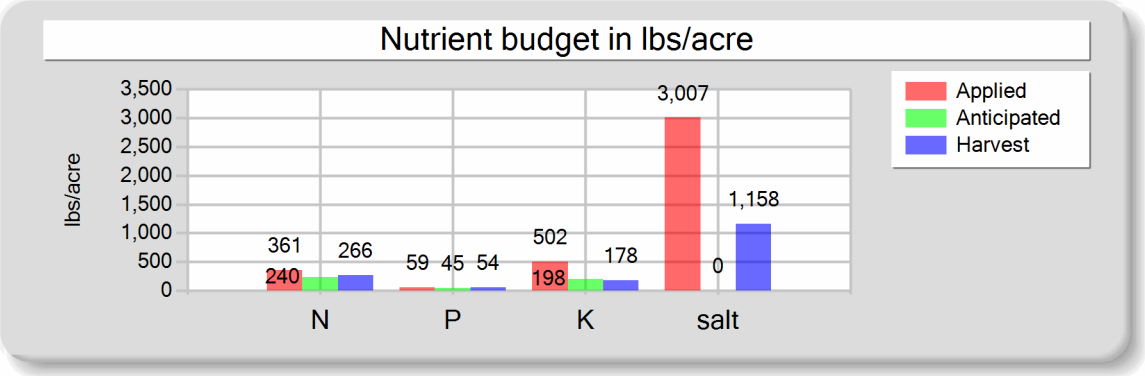
| Fresh water applied | |
|---|---|
| 2,929,725.00 | *gallons* |
| 107.89 | *acre-inches* |
| 10.79 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 684,000.00 | *gallons* |
| 25.19 | *acre-inches* |
| 2.52 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 2 - 05/30/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/30/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 354.24 | 59.11 | 502.34 | 2,712.87 |
| Fresh water | 0.00 | 0.00 | 0.00 | 294.23 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 361.24 | 59.11 | 502.34 | 3,007.09 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 265.86 | 53.60 | 177.96 | 1,157.79 |
| Nutrient balance | 95.38 | 5.51 | 324.38 | 1,849.31 |
| Applied to removed ratio | 1.36 | 1.10 | 2.82 | 2.60 |

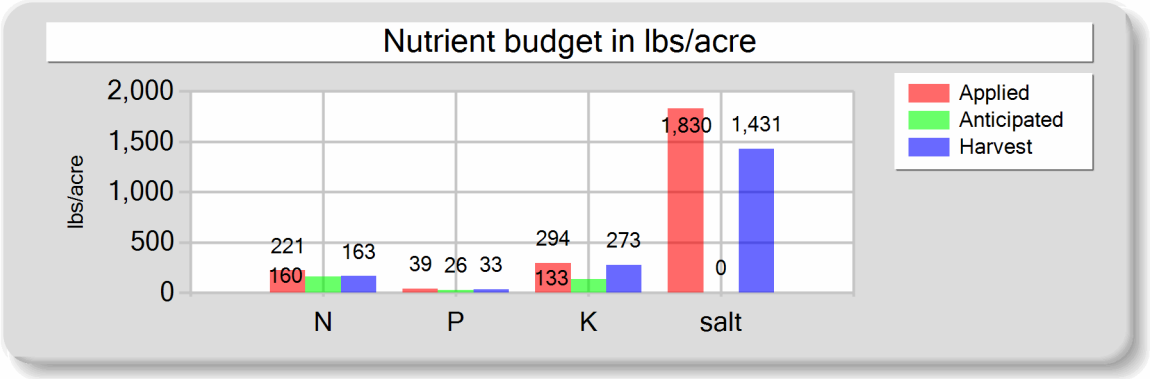
| Fresh water applied | |
|---|---|
| 11,752,575.00 | *gallons* |
| 432.81 | *acre-inches* |
| 43.28 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,080,000.00 | *gallons* |
| 39.77 | *acre-inches* |
| 3.98 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 3 - 12/08/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 12/08/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 213.53 | 39.49 | 293.98 | 1,754.60 |
| Fresh water | 0.00 | 0.00 | 0.00 | 75.03 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 220.53 | 39.49 | 293.98 | 1,829.63 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 163.15 | 32.95 | 273.25 | 1,430.56 |
| Nutrient balance | 57.38 | 6.54 | 20.72 | 399.08 |
| Applied to removed ratio | 1.35 | 1.20 | 1.08 | 1.28 |

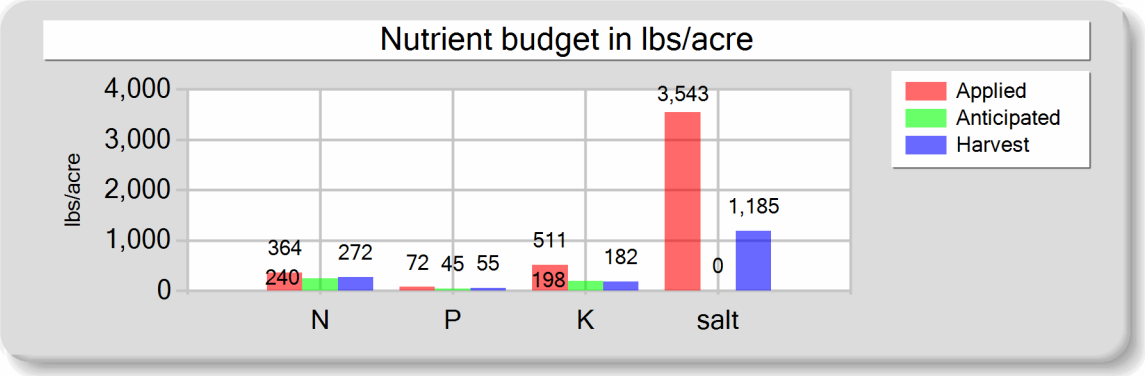
| Fresh water applied | |
|---|---|
| 5,994,150.00 | *gallons* |
| 220.74 | *acre-inches* |
| 11.04 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,260,000.00 | *gallons* |
| 46.40 | *acre-inches* |
| 2.32 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 3 - 05/15/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/15/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 356.73 | 72.48 | 510.51 | 3,277.73 |
| Fresh water | 0.00 | 0.00 | 0.00 | 265.56 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 363.73 | 72.48 | 510.51 | 3,543.29 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 272.09 | 54.86 | 182.13 | 1,184.93 |
| Nutrient balance | 91.64 | 17.62 | 328.39 | 2,358.37 |
| Applied to removed ratio | 1.34 | 1.32 | 2.80 | 2.99 |

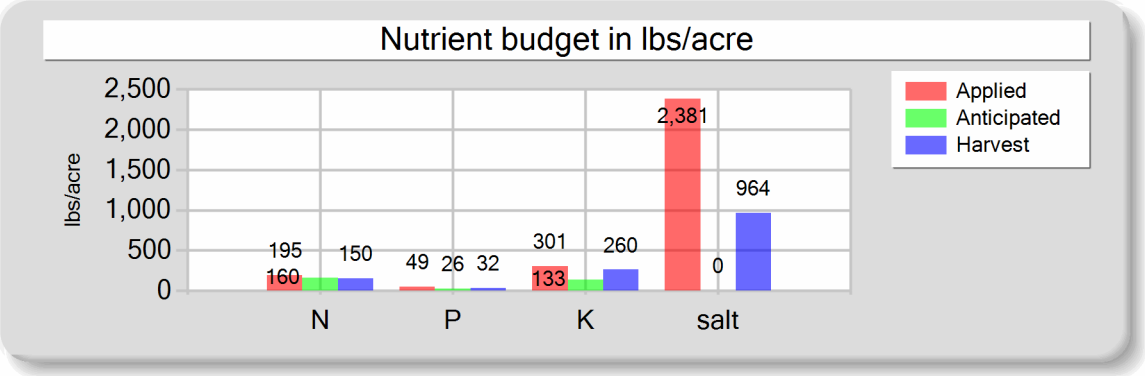
| Fresh water applied | |
|---|---|
| 21,215,250.00 | *gallons* |
| 781.29 | *acre-inches* |
| 39.06 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,367,000.00 | *gallons* |
| 87.17 | *acre-inches* |
| 4.36 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 5 - 12/08/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 12/08/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 188.05 | 49.03 | 301.35 | 2,304.04 |
| Fresh water | 0.00 | 0.00 | 0.00 | 76.87 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 195.05 | 49.03 | 301.35 | 2,380.91 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 149.54 | 31.88 | 259.60 | 964.02 |
| Nutrient balance | 45.51 | 17.15 | 41.75 | 1,416.89 |
| Applied to removed ratio | 1.30 | 1.54 | 1.16 | 2.47 |

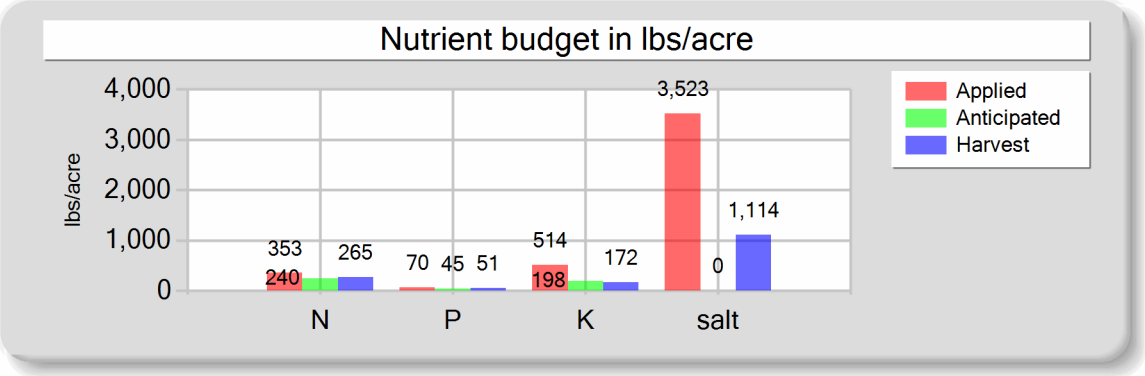
| Fresh water applied | |
|---|---|
| 5,219,625.00 | *gallons* |
| 192.22 | *acre-inches* |
| 11.31 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,431,000.00 | *gallons* |
| 52.70 | *acre-inches* |
| 3.10 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 5 - 05/15/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 05/15/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 345.72 | 69.84 | 514.14 | 3,242.33 |
| Fresh water | 0.00 | 0.00 | 0.00 | 280.19 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 352.72 | 69.84 | 514.14 | 3,522.52 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 265.21 | 51.26 | 171.61 | 1,114.33 |
| Nutrient balance | 87.51 | 18.58 | 342.54 | 2,408.19 |
| Applied to removed ratio | 1.33 | 1.36 | 3.00 | 3.16 |

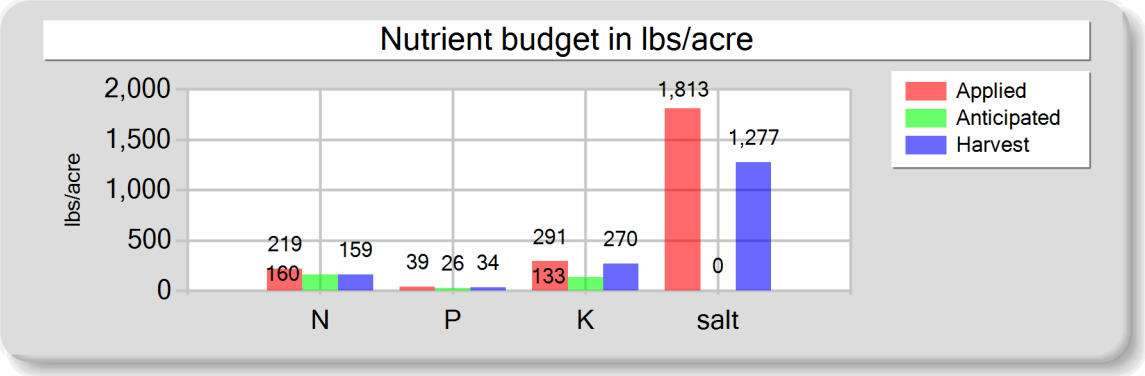
| Fresh water applied | |
|---|---|
| 19,026,375.00 | *gallons* |
| 700.68 | *acre-inches* |
| 41.22 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,106,000.00 | *gallons* |
| 77.56 | *acre-inches* |
| 4.56 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 6 - 12/08/2022: Oats, silage-soft dough

Field name: Field 6 Crop: Oats, silage-soft dough Plant date: 12/08/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 212.00 | 39.02 | 291.33 | 1,732.33 |
| Fresh water | 0.00 | 0.00 | 0.00 | 80.47 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 219.00 | 39.02 | 291.33 | 1,812.80 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 159.19 | 33.89 | 269.52 | 1,276.68 |
| Nutrient balance | 59.81 | 5.14 | 21.80 | 536.13 |
| Applied to removed ratio | 1.38 | 1.15 | 1.08 | 1.42 |

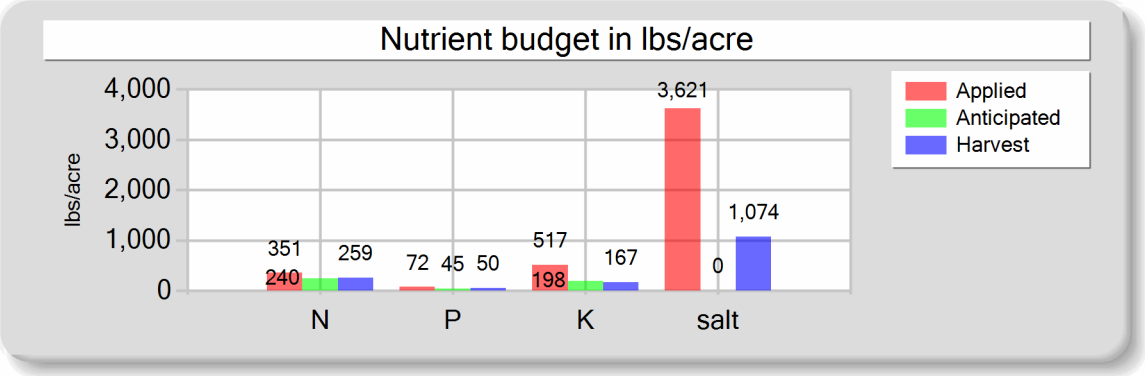
| Fresh water applied | |
|---|---|
| 3,535,875.00 | *gallons* |
| 130.21 | *acre-inches* |
| 11.84 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 684,000.00 | *gallons* |
| 25.19 | *acre-inches* |
| 2.29 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 6 - 05/15/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 05/15/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 344.41 | 71.82 | 516.59 | 3,339.65 |
| Fresh water | 0.00 | 0.00 | 0.00 | 281.27 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 351.41 | 71.82 | 516.59 | 3,620.92 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 258.74 | 50.43 | 166.65 | 1,074.43 |
| Nutrient balance | 92.67 | 21.38 | 349.94 | 2,546.49 |
| Applied to removed ratio | 1.36 | 1.42 | 3.10 | 3.37 |

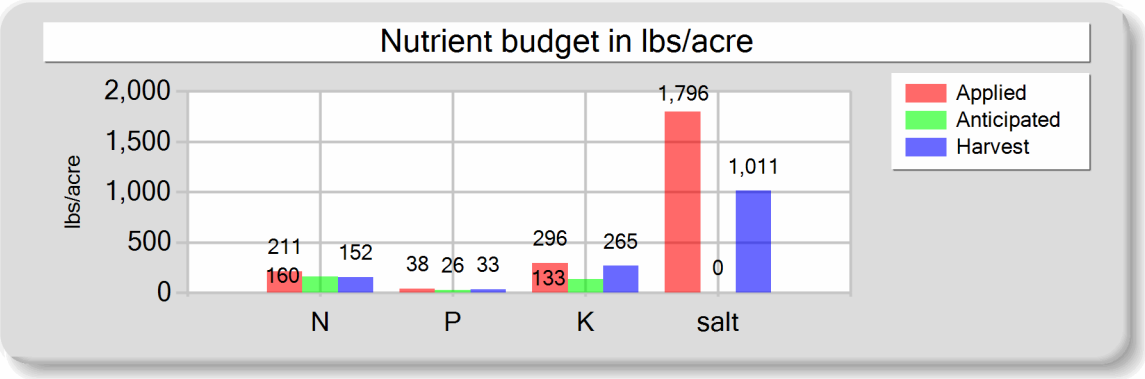
| Fresh water applied | |
|---|---|
| 12,358,725.00 | *gallons* |
| 455.13 | *acre-inches* |
| 41.38 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,395,000.00 | *gallons* |
| 51.37 | *acre-inches* |
| 4.67 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 7 - 12/08/2022: Oats, silage-soft dough

Field name: Field 7 Crop: Oats, silage-soft dough Plant date: 12/08/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 204.18 | 37.52 | 296.33 | 1,732.62 |
| Fresh water | 0.00 | 0.00 | 0.00 | 63.51 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 211.18 | 37.52 | 296.33 | 1,796.13 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 152.05 | 32.69 | 265.32 | 1,011.12 |
| Nutrient balance | 59.13 | 4.83 | 31.00 | 785.01 |
| Applied to removed ratio | 1.39 | 1.15 | 1.12 | 1.78 |

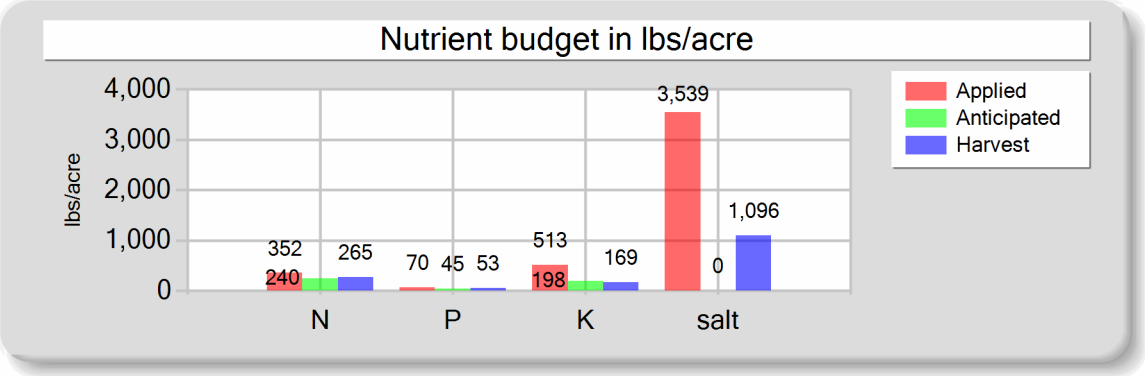
| Fresh water applied | |
|---|---|
| 7,610,550.00 | *gallons* |
| 280.27 | *acre-inches* |
| 9.34 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,025,000.00 | *gallons* |
| 74.57 | *acre-inches* |
| 2.49 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 7 - 05/15/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 05/15/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 344.67 | 70.08 | 513.47 | 3,254.83 |
| Fresh water | 0.00 | 0.00 | 0.00 | 283.83 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 351.67 | 70.08 | 513.47 | 3,538.65 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 265.17 | 52.60 | 168.74 | 1,095.74 |
| Nutrient balance | 86.50 | 17.49 | 344.73 | 2,442.91 |
| Applied to removed ratio | 1.33 | 1.33 | 3.04 | 3.23 |

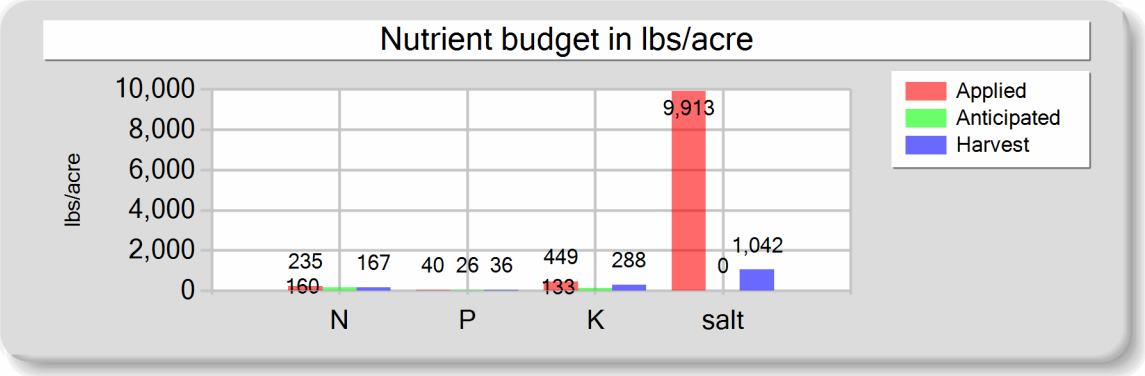
| Fresh water applied | |
|---|---|
| 34,011,750.00 | *gallons* |
| 1,252.54 | *acre-inches* |
| 41.75 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 3,726,000.00 | *gallons* |
| 137.22 | *acre-inches* |
| 4.57 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## HR-South - 12/08/2022: Oats, silage-soft dough

Field name: HR-South | Crop: Oats, silage-soft dough | Plant date: 12/08/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 127.27 | 40.30 | 145.30 | 8,081.85 |
| Process wastewater | 100.26 | 0.00 | 303.77 | 1,711.74 |
| Fresh water | 0.00 | 0.00 | 0.00 | 119.88 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 234.53 | 40.30 | 449.08 | 9,913.47 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 167.25 | 35.55 | 287.63 | 1,042.26 |
| Nutrient balance | 67.28 | 4.75 | 161.44 | 8,871.20 |
| Applied to removed ratio | 1.40 | 1.13 | 1.56 | 9.51 |

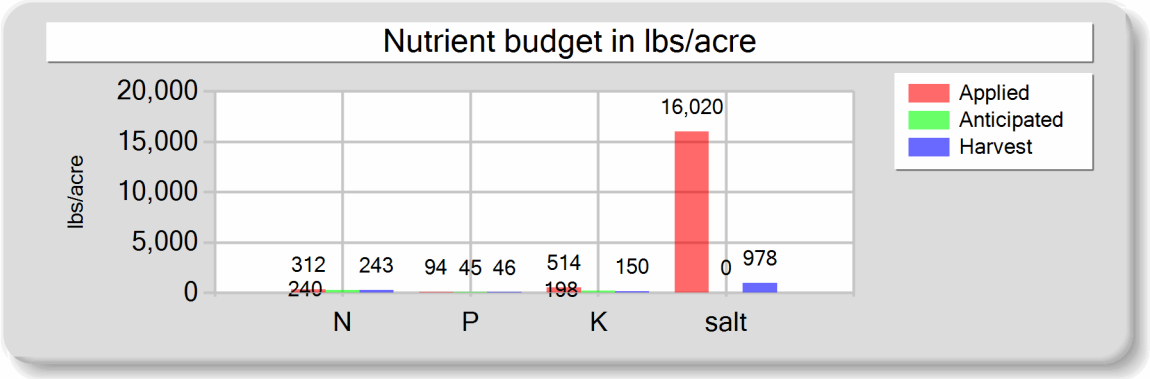
| Fresh water applied | |
|---|---|
| 39,265,050.00 | *gallons* |
| 1,446.00 | *acre-inches* |
| 17.63 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 4,406,644.00 | *gallons* |
| 162.28 | *acre-inches* |
| 1.98 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## HR-South - 05/15/2023: Corn, silage

Field name: HR-South | Crop: Corn, silage | Plant date: 05/15/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 202.04 | 94.39 | 303.74 | 14,078.81 |
| Process wastewater | 103.41 | 0.00 | 210.20 | 1,662.15 |
| Fresh water | 0.00 | 0.00 | 0.00 | 279.24 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 312.45 | 94.39 | 513.94 | 16,020.20 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 243.44 | 45.77 | 149.81 | 977.91 |
| Nutrient balance | 69.02 | 48.62 | 364.13 | 15,042.28 |
| Applied to removed ratio | 1.28 | 2.06 | 3.43 | 16.38 |

| Fresh water applied | |
|---|---|
| 91,461,300.00 | *gallons* |
| 3,368.21 | *acre-inches* |
| 41.08 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 3,798,138.00 | *gallons* |
| 139.87 | *acre-inches* |
| 1.71 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

NUTRIENT ANALYSES

## A. MANURE ANALYSES

### SMI - WJI

Sample and source description: SMI - WJI

Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 33.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 12,000.00 | 3,800.00 | 13,700.00 | | | | | | | 76.20 |
| **DL** | 0.10 | 0.10 | 0.10 | | | | | | | 10.00 |

### Solid Manure

Sample and source description: Solid Manure

Sample date: 05/15/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 22.9 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 12,100.00 | 4,400.00 | 6,100.00 | | | | | | | |
| **DL** | 100.00 | 100.00 | 100.00 | | | | | | | |

### Solid Manure

Sample and source description: Solid Manure

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 5.9 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 18,900.00 | 7,100.00 | 19,900.00 | | | | | | | |
| **DL** | 100.00 | 100.00 | 100.00 | | | | | | | |

SMI - WJI

Sample and source description: SMI - WJI

Sample date: 10/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 18.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 4,100.00 | 2,800.00 | 14,700.00 | | | | | | | 88.10 |
| **DL** | 0.10 | 0.10 | 0.10 | | | | | | | 10.00 |

## B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 555.00 | 393.00 | 0.00 | 2.00 | 80.90 | 701.00 | | | | | | | | 7,190.00 | 3,400 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

LWI - WJF

Sample and source description: LWI - WJF

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 314.00 | 59.40 | 0.00 | 0.00 | 0.00 | 961.00 | | | | | | | | 5,540.00 | 4,120 |
| **DL** | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | | | | | | | | 10.00 | 10 |

## LWI - WJF

Sample and source description: LWI - WJF

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 627.00 | 32.80 | 0.00 | 0.00 | 0.00 | 996.00 | | | | | | | | 7,730.00 | 6,340 |
| **DL** | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | | | | | | | | 10.00 | 10 |

## Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 457.00 | 356.00 | 0.00 | 1.10 | 63.40 | 624.00 | | | | | | | | 7,130.00 | 2,870 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

## Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 267.00 | 230.00 | 0.00 | 0.70 | 69.80 | 429.00 | | | | | | | | 5,330.00 | 3,280 |
| **DL** | 1.00 | 0.50 | 1.00 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

## Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 536.00 | 371.00 | 0.00 | 0.10 | 100.00 | 739.00 | | | | | | | | 7,290.00 | 4,350 |
| **DL** | 1.00 | 0.50 | 1.00 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

LWI - WJF

Sample and source description: LWI - WJF

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 123.00 | 27.70 | 0.00 | 0.00 | 0.00 | 362.00 | | | | | | | | 3,010.00 | 3,480 |
| **DL** | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | | | | | | | | 10.00 | 10 |

## C. FRESH WATER ANALYSES

Domestic Well - 1-B 9436 Griffith

1-B 9436 Griffith

Sample description: 1-B 9436 Griffith

Sample date: 11/15/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 35.50 | 0.00 | 35.50 | | | | | | | | 856.00 | |
| **DL** | 1.00 | 1.00 | 0.10 | | | | | | | | 10.00 | |

Domestic Well - 2-H 9436 Griffith

2-H 9436 Griffith

Sample description: 2-H 9436 Griffith

Sample date: 11/15/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 35.90 | 0.00 | 35.90 | | | | | | | | 859.00 | |
| **DL** | 0.10 | 1.00 | 0.10 | | | | | | | | 10.00 | |

Domestic Well - 4-17573 Schendel

Domestic Well - 4-17573 Schendel

4-17573 Schendel

Sample description: 4-17573 Schendel

Sample date: 11/15/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 73.60 | 0.00 | 73.60 | | | | | | | | 1,780.00 | |
| **DL** | 0.10 | 1.00 | 0.10 | | | | | | | | 10.00 | |

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/24/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 0.00 | 0.00 | 0.00 | | | | | | | | 32.00 | 30 |
| **DL** | 1.00 | 1.00 | 1.00 | | | | | | | | 10.00 | 10 |

## D. SOIL ANALYSES

Field 2

2 & 3

Sample and source description: 2 & 3

Sample date: 11/29/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 104.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field 2

### 2 & 3

Sample and source description: 2 & 3

Sample date: 11/29/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 104.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field 5

### 5

Sample and source description: 5

Sample date: 11/29/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 77.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field 6

### 6

Sample and source description: 6

Sample date: 11/29/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 57.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field 7

Field 7

7

Sample and source description: 7

Sample date: 11/29/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 85.00 | | | | |
| **DL** | | | 2.00 | | | | |

HR-South

HR-South B

Sample and source description: HR-South B

Sample date: 01/29/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 85.00 | | | | |
| **DL** | | | 2.00 | | | | |

HR-South A

Sample and source description: HR-South A

Sample date: 11/29/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 69.00 | | | | |
| **DL** | | | 2.00 | | | | |

## E. PLANT TISSUE ANALYSES

Field 2 - 12/08/2022: Oats, silage-soft dough

Field 2 - 12/08/2022: Oats, silage-soft dough

2 & 3

Sample and source description: 2 & 3

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 20,300.00 | 4,100.00 | 34,000.00 | | 17.80 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Field 2 - 05/30/2023: Corn, silage

2 & 3

Sample and source description: 2 & 3

Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,400.00 | 2,500.00 | 8,300.00 | | 5.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Field 3 - 12/08/2022: Oats, silage-soft dough

2 & 3

Sample and source description: 2 & 3

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 20,300.00 | 4,100.00 | 34,000.00 | | 17.80 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Field 3 - 05/15/2023: Corn, silage

2 & 3

Sample and source description: 2 & 3

Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,400.00 | 2,500.00 | 8,300.00 | | 5.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Field 5 - 12/08/2022: Oats, silage-soft dough

5

Sample and source description: 5

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 19,700.00 | 4,200.00 | 34,200.00 | | 12.70 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Field 5 - 05/15/2023: Corn, silage

5

Sample and source description: 5

Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,900.00 | 2,300.00 | 7,700.00 | | 5.00 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Field 6 - 12/08/2022: Oats, silage-soft dough

6

Sample and source description: 6

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 20,200.00 | 4,300.00 | 34,200.00 | | 16.20 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Field 6 - 05/15/2023: Corn, silage

6

Sample and source description: 6

Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.5 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,800.00 | 2,300.00 | 7,600.00 | | 4.90 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Field 7 - 12/08/2022: Oats, silage-soft dough

7

Sample and source description: 7

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 20,000.00 | 4,300.00 | 34,900.00 | | 13.30 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 7 - 05/15/2023: Corn, silage

### 7

Sample and source description: 7

Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,100.00 | 2,400.00 | 7,700.00 | | 5.00 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## HR-South - 12/08/2022: Oats, silage-soft dough

### HR-South A

Sample and source description: HR-South A

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 20,700.00 | 4,400.00 | 35,600.00 | | 12.90 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

### HR-South B

Sample and source description: HR-South B

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 20,200.00 | 3,900.00 | 34,500.00 | | 18.00 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## HR-South - 05/15/2023: Corn, silage

HR-South - 05/15/2023: Corn, silage

HR-South A

Sample and source description: HR-South A

Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.5 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,700.00 | 2,200.00 | 7,200.00 | | 4.70 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

HR-South B

Sample and source description: HR-South B

Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.5 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,700.00 | 2,100.00 | 7,000.00 | | 4.50 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## F. SUBSURFACE (TILE) DRAINAGE ANALYSES

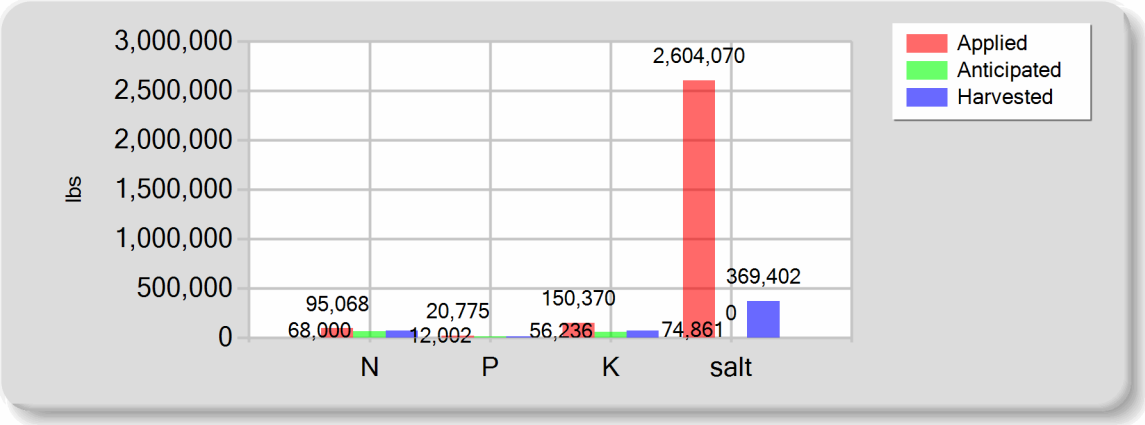
*No subsurface (tile) drainage analyses entered.*

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

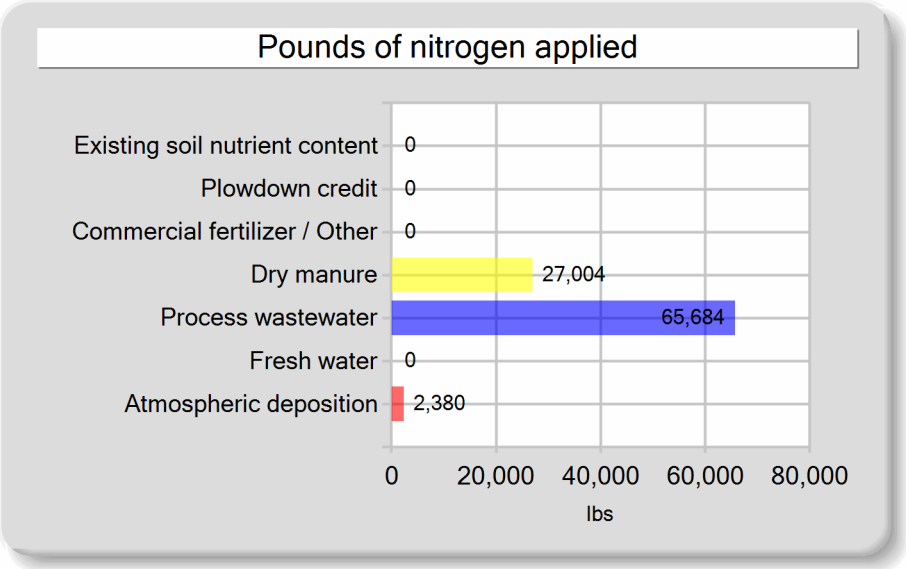
## A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

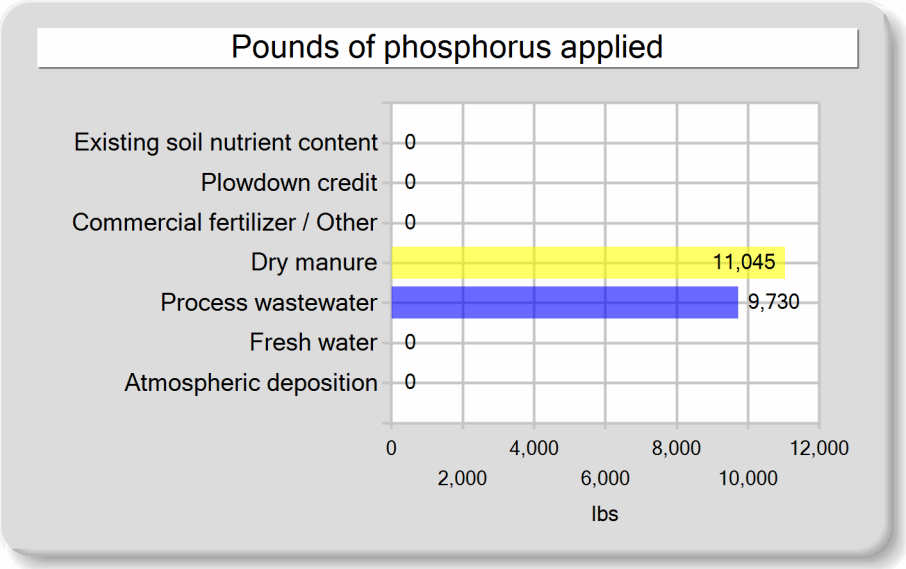
| | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 27,003.96 | 11,044.86 | 36,821.49 | 1,817,173.80 |
| Process wastewater | 65,683.74 | 9,729.70 | 113,548.95 | 723,211.86 |
| Fresh water | 0.00 | 0.00 | 0.00 | 63,684.27 |
| Atmospheric deposition | 2,380.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 95,067.70 | 20,774.56 | 150,370.44 | 2,604,069.93 |
| Anticipated crop nutrient removal | 68,000.00 | 12,002.00 | 56,236.00 | 0.00 |
| Actual crop nutrient removal | 70,967.21 | 14,216.39 | 74,860.67 | 369,401.76 |
| Nutrient balance | 24,100.49 | 6,558.18 | 75,509.77 | 2,234,668.17 |
| Applied to removed ratio | 1.34 | 1.46 | 2.01 | 7.05 |

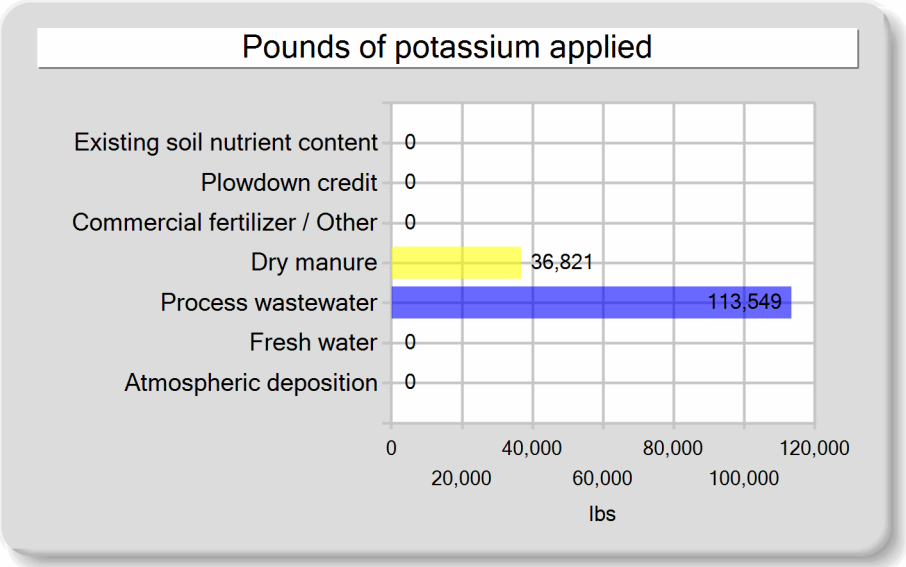
## B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

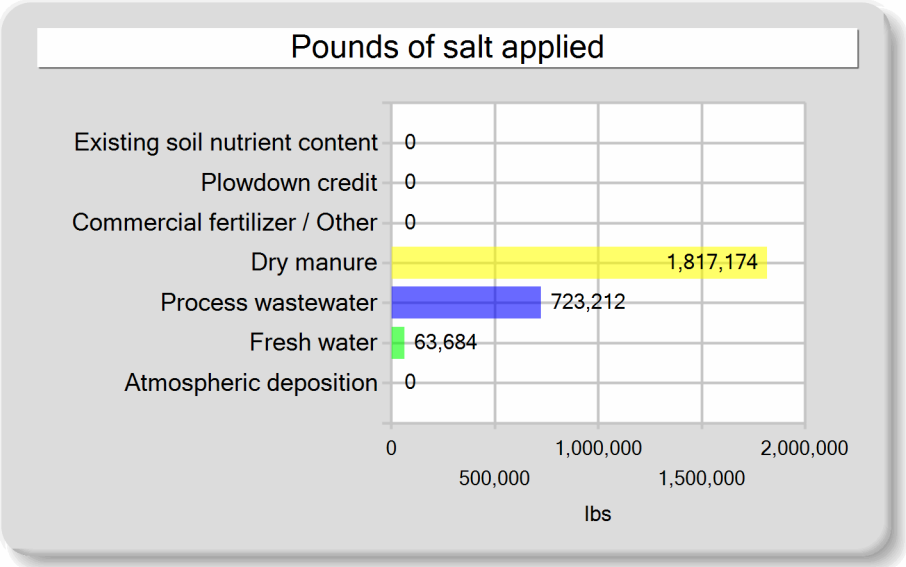


## C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE



## EXCEPTION REPORTING

### A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

*No manure or process wastewater discharges occurred during the reporting period.*

### B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

*No stormwater discharges occurred during the reporting period.*

### C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

*No land application area to surface water discharges occurred during the reporting period.*

## NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

### A. NUTRIENT MANAGEMENT PLAN STATEMENTS

| | |
|---|---|
| Was the facility's NMP updated in the reporting period? | Yes |
| Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |
| Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |

### B. EXPORT AGREEMENT STATEMENT

| | |
|---|---|
| Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? | No |

ADDITIONAL NOTES

**A. NOTES**

*No notes entered for this annual report.*

CERTIFICATION

**A. OWNER AND/OR OPERATOR CERTIFICATION**

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| | |
|---|---|
| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
| Paul Caetano | John Vitorino |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| 06/21/2024 | 06/24/2024 |
| DATE | DATE |

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Vitorino Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 08:53

## Samples in this Report

| Lab ID | Sample | Matrix | Sampled By | Crop | Date Sampled |
|---|---|---|---|---|---|
| 23K0888-01 | 1-B 9436 Griffith | Ag Water | T. Madruga | | 11/15/2023 15:00 |
| 23K0888-02 | 2-H 9436 Griffith | Ag Water | T. Madruga | | 11/15/2023 15:30 |
| 23K0888-03 | 4-17573 Schendel | Ag Water | T. Madruga | | 11/15/2023 16:00 |

Default Cooler Temperature on Receipt °C: -0.2
Containers Intact
COC/Labels Agree
Received On Ice

## Notes and Definitions

| Item | Definition |
|---|---|
| H | Hold Time Exceeded |
| MCL | Drinking Water Maximum Contaminant Level |
| ND | Analyte NOT DETECTED at or above the reporting limit. |
| NES | Not Enough Sample |
| * | Not Taken |
| RPD | Relative Percent Difference |
| %REC | Percent Recovery |
| Source | Sample that was matrix spiked or duplicated. |

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Ranch: Vitorino Dairy

Received: 11/16/2023 15:35
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## Sample Results

**Sample: 1-B 9436 Griffith**
**23K0888-01 (Water)**

Sampled: 11/15/2023 15:00
Sampled By: T. Madruga

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **0.86** | mmhos/cm | 0.01 | 1 | | 11/20/23 12:54 | SM 2510 B | | BEK0615 |
| **Electrical Conductivity umhos** | **856** | umhos/cm | 10.0 | 1 | | 11/20/23 12:54 | SM 2510 B | | BEK0615 |
| Ammonia (as N) | ND | mg/L | 0.00 | 1 | | 11/15/23 15:00 | Field | | BEK0549 |
| **Nitrate Nitrogen as NO3N** | **35.5** | mg/L | 0.1 | 1 | 10 | 11/16/23 19:48 | EPA 300.0 | | BEK0572 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 11/20/23 12:54 | SM 4500-H+ | H | BEK0615 |
| **pH** | **7.6** | units | 1.0 | 1 | | 11/20/23 12:54 | SM 4500-H+ | H | BEK0615 |

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8119 Ellington Ave
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Account# 00-0019840
Account Manager: Ben Nydam
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Ranch: Vitorino Dairy

Received: 11/16/2023 15:35
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## Sample Results
**(Continued)**

**Sample: 2-H 9436 Griffith**
**23K0888-02 (Water)**

Sampled: 11/15/2023 15:30
Sampled By: T. Madruga

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **0.86** | mmhos/cm | 0.01 | 1 | | 11/20/23 12:55 | SM 2510 B | | BEK0615 |
| **Electrical Conductivity umhos** | **859** | umhos/cm | 10.0 | 1 | | 11/20/23 12:55 | SM 2510 B | | BEK0615 |
| Ammonia (as N) | ND | mg/L | 0.00 | 1 | | 11/15/23 15:30 | Field | | BEK0549 |
| **Nitrate Nitrogen as NO3N** | **35.9** | mg/L | 0.1 | 1 | 10 | 11/16/23 20:08 | EPA 300.0 | | BEK0572 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 11/20/23 12:55 | SM 4500-H+ | H | BEK0615 |
| **pH** | **7.7** | units | 1.0 | 1 | | 11/20/23 12:55 | SM 4500-H+ | H | BEK0615 |

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Ranch: Vitorino Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 08:53

## Sample Results
**(Continued)**

**Sample: 4-17573 Schendel**
**23K0888-03 (Water)**

Sampled: 11/15/2023 16:00
Sampled By: T. Madruga

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **1.78** | mmhos/cm | 0.01 | 1 | | 11/20/23 12:56 | SM 2510 B | | BEK0615 |
| **Electrical Conductivity umhos** | **1780** | umhos/cm | 10.0 | 1 | | 11/20/23 12:56 | SM 2510 B | | BEK0615 |
| Ammonia (as N) | ND | mg/L | 0.00 | 1 | | 11/15/23 16:00 | Field | | BEK0549 |
| **Nitrate Nitrogen as NO3N** | **73.6** | mg/L | 0.1 | 1 | 10 | 11/16/23 20:29 | EPA 300.0 | | BEK0572 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 11/20/23 12:56 | SM 4500-H+ | H | BEK0615 |
| **pH** | **7.4** | units | 1.0 | 1 | | 11/20/23 12:56 | SM 4500-H+ | H | BEK0615 |

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Ranch: Vitorino Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 08:53

## Quality Control

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0572*** | | | | | | | | | | |
| **Blank (BEK0572-BLK1)** | | | | | Prepared & Analyzed: 11/16/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEK0572-BLK2)** | | | | | Prepared & Analyzed: 11/16/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEK0572-BLK3)** | | | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **LCS (BEK0572-BS1)** | | | | | Prepared & Analyzed: 11/16/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.3 | | 0.1 | mg/L | 5.000 | | 107 | 90-110 | | |
| **LCS (BEK0572-BS2)** | | | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.0 | | 0.1 | mg/L | 5.000 | | 99.1 | 90-110 | | |
| **Duplicate (BEK0572-DUP1)** | | **Source: 23K0900-03** | | | Prepared & Analyzed: 11/16/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 4.3 | | 0.1 | mg/L | | 4.3 | | | 0.0693 | 10 |
| **Duplicate (BEK0572-DUP2)** | | **Source: 23K0943-01** | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 1.4 | | 0.1 | mg/L | | 1.4 | | | 0.215 | 10 |
| **Matrix Spike (BEK0572-MS1)** | | **Source: 23K0900-03** | | | Prepared & Analyzed: 11/16/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 9.4 | | 0.1 | mg/L | 5.000 | 4.3 | 102 | 90-110 | | |
| **Matrix Spike (BEK0572-MS2)** | | **Source: 23K0943-01** | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 6.5 | | 0.1 | mg/L | 5.000 | 1.4 | 102 | 90-110 | | |
| **Reference (BEK0572-SRM1)** | | | | | Prepared & Analyzed: 11/16/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.0 | | | mg/L | 10.00 | | 99.8 | 90-110 | | |
| **Reference (BEK0572-SRM2)** | | | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.0 | | | mg/L | 10.00 | | 100 | 90-110 | | |
| **Reference (BEK0572-SRM3)** | | | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.1 | | | mg/L | 10.00 | | 101 | 90-110 | | |

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Ranch: Vitorino Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 08:53

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0615*** | | | | | | | | | | |
| **Blank (BEK0615-BLK1)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| pH | 5.7 | | 1.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| **Blank (BEK0615-BLK2)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 6.4 | | 1.0 | units | | | | | | |
| **Blank (BEK0615-BLK3)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| pH | 6.0 | | 1.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| **Duplicate (BEK0615-DUP1)** | | **Source: 23K0888-03** | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 1.78 | | 0.01 | mmhos/cm | | 1.78 | | | 0.477 | 10 |
| pH | 7.4 | | 1.0 | units | | 7.4 | | | 0.271 | 10 |
| Electrical Conductivity umhos | 1780 | | 10.0 | umhos/cm | | 1780 | | | 0.477 | 10 |
| **Duplicate (BEK0615-DUP2)** | | **Source: 23K0892-03** | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 0.84 | | 0.01 | mmhos/cm | | 0.84 | | | 0.262 | 10 |
| Electrical Conductivity umhos | 841 | | 10.0 | umhos/cm | | 839 | | | 0.262 | 10 |
| pH | 7.6 | | 1.0 | units | | 7.6 | | | 0.00 | 10 |
| **Reference (BEK0615-SRM1)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 550 | | | umhos/cm | 538.0 | | 102 | 90-110 | | |
| **Reference (BEK0615-SRM2)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| pH | 7.6 | | | units | 7.520 | | 101 | 67021-101.32 | | |
| **Reference (BEK0615-SRM3)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 1020 | | | umhos/cm | 1000 | | 102 | 90-110 | | |
| Electrical Conductivity umhos | 1020 | | | umhos/cm | 1000 | | 102 | 90-110 | | |
| **Reference (BEK0615-SRM4)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |
| Electrical Conductivity umhos | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |
| **Reference (BEK0615-SRM5)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 1030 | | | umhos/cm | 1000 | | 103 | 90-110 | | |

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Vitorino Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 08:53

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0615 (Continued)*** | | | | | | | | | | |
| **Reference (BEK0615-SRM5)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity umhos | 1030 | | | umhos/cm | 1000 | | 103 | 90-110 | | |
| **Reference (BEK0615-SRM6)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |
| **Reference (BEK0615-SRM7)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |
| **Reference (BEK0615-SRM8)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |

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11/16/23 15:35 23K0888

# DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 3 No. of Bottles: 3

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

**Name:** R & J Avila Inc
**Address:** 8119 Ellington Avenue
**City:** Hilmar **State:** CA **Zip:** 95324
**Telephone:** (209) 669-7919 **Fax:** (209) 634-4604
**Cell/Email:** 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com

**COPY TO:**

**REQUESTED BY:** Rui & John Avila
**PROJECT** Vitorino Dairy
**ID CROP:** N/A

[X] Copy of Chain [X] QA/QC Documents
**Sampled By:** T. Madruga

**Other:**

**Analysis and Bottles Required: (Please indicate Analysis)**

**( X ) DWW1: EC, pH, $NO_3$-N (NH4-N Field Test)**
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus $SO_4$, $CO_3$, $HCO_3$, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, $NO_3$-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, $NO_3$-N, $NH_4$-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus $SO_4$, $CO_3$, $HCO_3$, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

| | Description of Samples | Date Sampled | Time Sampled | Rec'd Temp °C | Field $NH_4$-N |
|---|---|---|---|---|---|
| 1 | 1-B 9436 Griffith | 11/15/2023 | 1500 | -.2 | ND |
| 2 | 2-H 9436 Griffith | 11/15/2023 | 1530 | .5 | ND |
| 3 | 4-17573 Schendel | 11/15/2023 | 1600 | .3 | ND |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |
| 7 | | | | | |
| 8 | | | | | |
| 9 | | | | | |
| 10 | | | | | |

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

**CHAIN OF CUSTODY**

| Carrier | Signature | Company | Received (Date/Time) | Relinquished (Date/Time) |
|---|---|---|---|---|
| First | John Avila | R & J Avila, Inc. | 11/15/2023 @ 1600 | 11/16/2023 @9am |
| Second | JCE | Dellavalle Laboratory, Inc. | 11/16/2023 @9am | 11-16-23 3:25 |
| Third | ~~DLE~~ | DLA | 11-16-23 15:35 | |
| Fourth | | | | |

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal* ). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

***Invoicing Information:*** ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

| Amt Paid | Rec By | Check # | Date |
|---|---|---|---|
| | | | |

**Signature**

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/16/23 15:35 23K0888

**Shipping Information:** Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

**Shipping Container:** Ice Chest ☒ None ☐ Box ☐

**Refrigerant:** Wet Ice: ☒ Blue Ice: ☐ None: ☐

| Container(s) Received | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Plastics:** | | | | | | | | | | |
| 100 mL sterile plastic $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 250 mL unpreserved (White) | | | | | | | | | | |
| 250 mL $HNO_3$ (Red) | | | | | | | | | | |
| 250 mL $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 500 mL unpreserved (White) | | | | | | | | | | |
| 500 mL $HNO_3$ (Red) | | | | | | | | | | |
| 500 mL $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 1 L unpreserved (White) | 1 | 1 | 1 | | | | | | | |
| 1 L unpreserved (BOD) (Purple) | | | | | | | | | | |
| 1 L $HNO_3$ (Red) | | | | | | | | | | |
| 1 L $H_2SO_4$ (Yellow) | | | | | | | | | | |
| **VOA Vials in sets of 3:** | | | | | | | | | | |
| 40 mL VOA, HCl (Blue) | | | | | | | | | | |
| 40 mL VOA, $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 40 mL VOA, unpreserved (White) | | | | | | | | | | |
| **Glass:** | | | | | | | | | | |
| 250 mL AG unpreserved (White) | | | | | | | | | | |
| 250 mL AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 250 mL AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 250 mL AG $Na_2S_2O_3$ + MCAA | | | | | | | | | | |
| 500 mL glass unpreserved (White) | | | | | | | | | | |
| 500 mL AG HCl (Blue) | | | | | | | | | | |
| 1 L AG unpreserved (White) | | | | | | | | | | |
| 1 L AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 1 L AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 1 L AG HCl (Blue) | | | | | | | | | | |
| **Special:** | | | | | | | | | | |
| $Cr^{6+}$ w/Borate/HCO3/CO3 | | | | | | | | | | |
| Cyanide - 500 mL NaOH | | | | | | | | | | |
| Asbestos - 2 - 1L P wrapped in foil | | | | | | | | | | |
| 500 mL or 1 L **AP** $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 1 L **AG** or **P** NaOH + ZnAc (Sulfide) | | | | | | | | | | |
| Chlorite/Bromate 250 mL AG with EDA | | | | | | | | | | |
| **TTHM** | | | | | | | | | | |
| **HAA5** | | | | | | | | | | |
| 40 mL VOA, $Na_2S_2O_3$ + MCAA | | | | | | | | | | |